

WILL THEY STAY OR WILL THEY CHURN?

Gym Member Engagement & Retention Analysis

Analyzing member behavior, engagement, accessibility, and observed churn during the first two months of membership.

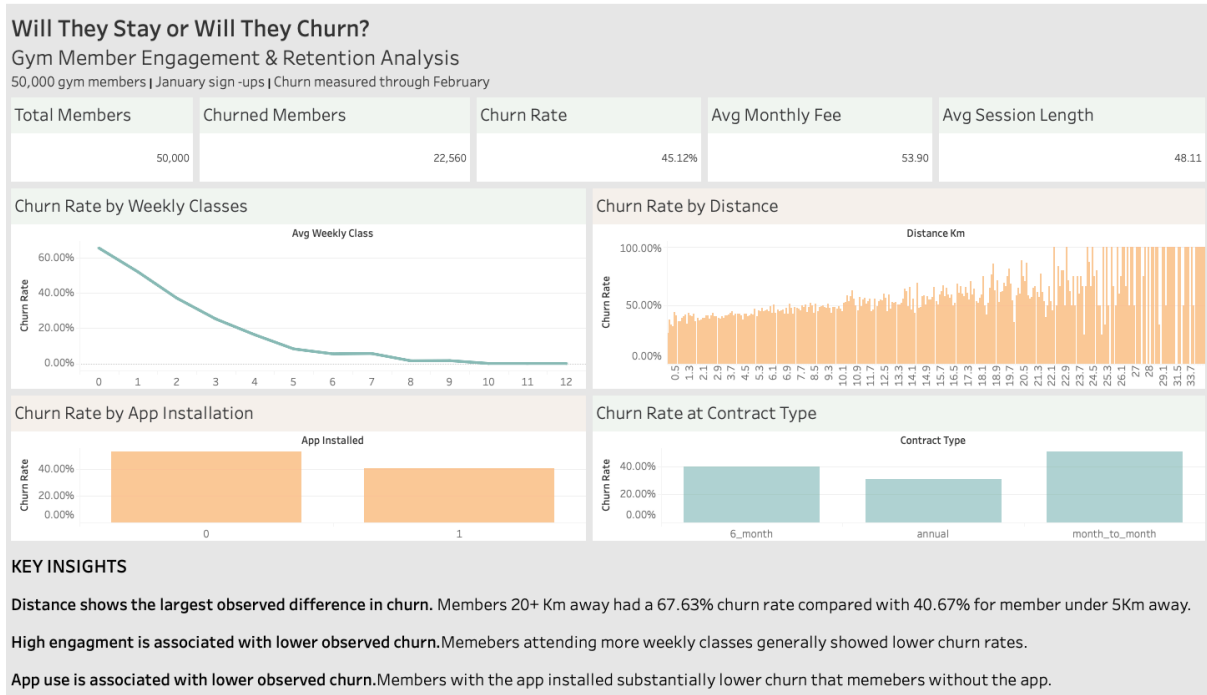
50,000 Members | January Sign-ups | Churn measured through February

Tools: SQL | Excel | Tableau

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Data Analytics Portfolio Project

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Executive Summary

Gym member retention varied substantially across accessibility and engagement behaviors during the first two months of membership. The analysis identified distance from the gym, weekly class participation, and average session length as the strongest observed relationships with churn, while demographic characteristics showed comparatively smaller differences.

A few key metrics from the analysis:

45.12%
of the 50,000 members analyzed churned by February.

67.63% vs. 40.67%
churn among members living 20+ km away versus members living under 5 km away.

65.33%
churn among members attending zero weekly classes.

59.78% vs. 33.80%
churn among members with sessions under 30 minutes versus 60+ minutes.

11.7%
of the variation in churn was explained by weekly class participation in the exploratory regression model, the strongest of the three regression models analyzed.

The findings suggest that **early engagement and accessibility are important areas for retention strategy**. Members showing limited participation may benefit from earlier engagement efforts, while distance-related barriers may warrant additional consideration when designing retention initiatives.

Approach

I used gym membership data to evaluate retention, identify patterns in member engagement, and investigate factors associated with churn. SQL was used for data preparation and exploratory analysis, Excel was used for statistical testing, and Tableau was used to communicate the findings through an interactive dashboard.

Business Problem

Gyms invest significant resources in acquiring new members, but early member churn can reduce long-term revenue and engagement.

A fitness business needs to understand:

- Which member behaviors are associated with churn?
- Which forms of engagement appear most important?
- Are accessibility factors related to retention?
- Can early member behavior provide useful signals for retention strategies?

Research Question

What factors are associated with gym member engagement and churn, and how can a fitness business use these insights to improve member retention?

Business Questions

1. How does member engagement relate to churn?
2. Is distance from the gym associated with churn?
3. Is weekly class participation associated with churn?
4. Is average session length associated with churn?
5. Are member characteristics such as contract type, age, sex, and primary goal meaningfully different across churn groups?

Dataset

50,000 gym membership records

The dataset included information about:

- Member demographics
- Distance from the gym
- Contract type
- Monthly membership fee
- Prior gym experience
- Joining with a friend
- Promotional activity
- Primary fitness goal
- Weekly visits
- Weekly classes
- Personal trainer usage
- App installation
- Average session duration
- Guest pass usage
- Locker rental
- Churn status

Data Preparation

SQL was used to:

- Inspect the database schema
- Check for missing values
- Check member ID uniqueness
- Validate categorical values
- Review numeric ranges
- Calculate churn rates
- Segment members
- Compare engagement behaviors
- Prepare analysis-ready data

SQL Chunk example:

```
SELECT
  COUNT(*) AS total_members,
  SUM(churned_by_febru) AS churned_members,
  ROUND(100.0 * SUM(churned_by_febru) / COUNT(*), 2 ) AS churn_rate
FROM users;
```

Metric	Result
Total Members	50,000
Churned Members	20,560
Churn Rate	45.12%
Retained Members	54.88%

Overall Member Retention

Of the 50,000 members analyzed, 20,560 churned by February, resulting in an overall churn rate of 45.12%.

Churn by Distance

Distance	Churn Rate
Under 5 km	40.67%
5–9.99 km	46.46%

10–19.99 km	54.74%
20+ km	67.63%

Insight

Churn increased substantially as distance from the gym increased. Members living 20+ km away had a 67.63% observed churn rate compared with 40.67% among members living under 5 km away, a difference of 26.96 percentage points.

Potential opportunity: Explore retention strategies for members who face greater geographic barriers to accessing the facility.

Weekly Class Participation



Weekly class participation showed a strong negative relationship with observed churn. Members attending zero weekly classes had a 65.33% churn rate, while churn generally decreased as weekly class participation increased.

Then statistical result:

Exploratory regression

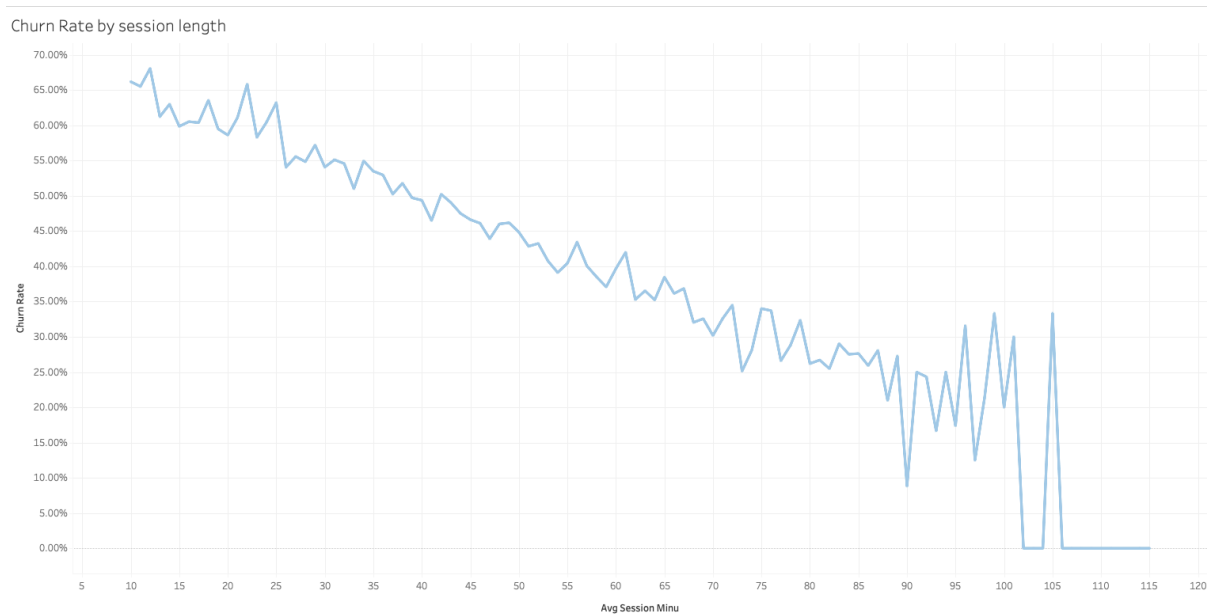
- $\beta = -0.1098$
- $R^2 = 0.117$
- $p < 0.001$

Interpretation

The regression showed a statistically significant negative association between weekly class participation and churn. The model explained approximately 11.7%

of the variation in the binary churn indicator, making it the strongest of the three exploratory regression models.

Average Session Length & Churn



Members with shorter average sessions showed higher observed churn. Members averaging under 30 minutes per session had a 59.78% churn rate compared with 33.80% among members averaging 60+ minutes.

Statistical result:

- $\beta = -0.00553$
- $R^2 = 0.0317$
- $p < 0.001$

Interpretation

Average session length was significantly and negatively associated with churn. However, the relatively low R^2 indicates that session length alone explains only a small portion of the variation in churn.

App Installation

Two-sample t-test found:

t = -27.43
df = 42,276
p = 2.613×10^{-164}

Interpretation

The test indicated a statistically significant difference in the mean churn indicator between members who did and did not have the gym app installed. Therefore, Members without the app showed substantially higher observed churn than members with the app installed.

Statistical Analysis

Hypothesis testing

Relationship	Method	Key Result	p-value
Distance → Churn	Linear regression	beta = +0.013, R ² = .013	< 0.001
Weekly Classes → Churn	Linear regression	beta = -0.1098, R ² = 0.117	< 0.001
Session Length → Churn	Linear regression	beta = -0.00553, R ² = 0.0317	< 0.001
App Installation → Churn	Welch t-test	t = -27.43	2.613×10^{-164}

Statistical note

Because churn is a binary outcome, the regression analyses should be interpreted as exploratory linear probability models rather than formal predictive churn models. Logistic regression would be a more appropriate modeling approach for future analysis.

Tableau Dashboard

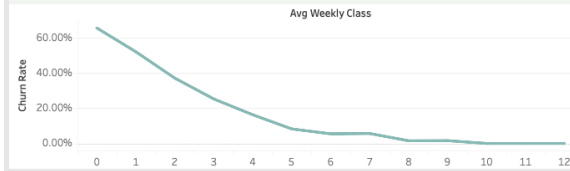
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Gym Member Engagement & Retention Analysis

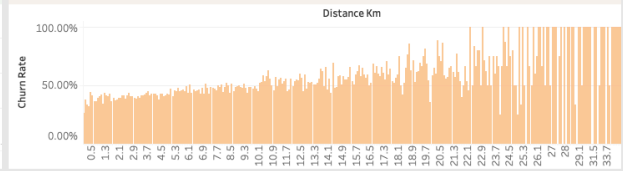
50,000 gym members | January sign-ups | Churn measured through February

Total Members	Churned Members	Churn Rate	Avg Monthly Fee	Avg Session Length
50,000	22,560	45.12%	53.90	48.11

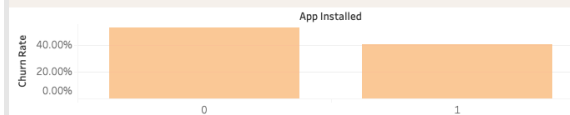
Churn Rate by Weekly Classes



Churn Rate by Distance



Churn Rate by App Installation



Churn Rate at Contract Type



KEY INSIGHTS

Distance shows the largest observed difference in churn. Members 20+ Km away had a 67.63% churn rate compared with 40.67% for member under 5Km away.

High engagement is associated with lower observed churn. Members attending more weekly classes generally showed lower churn rates.

App use is associated with lower observed churn. Members with the app installed substantially lower churn than members without the app.

Dashboard Highlights

The dashboard summarizes:

- Overall member and churn KPIs
- Churn by weekly class participation
- Churn by app installation
- Churn by contract type
- Churn by distance
- Average member/session metrics

[\[View Interactive Tableau Dashboard\]](#)

Recommendations

1. Prioritize early engagement

Members showing little or no weekly class participation had substantially higher observed churn.

Potential action:

Create an early-member engagement program that identifies members with low participation during their first few weeks and offers personalized outreach, class recommendations, or onboarding support.

2. Investigate accessibility barriers

Distance showed one of the largest observed differences in churn.

Potential action:

Consider targeted retention strategies for members living farther from the facility, such as flexible scheduling, virtual resources, or incentives that make regular visits more convenient.

3. Encourage meaningful gym engagement

Shorter average sessions were associated with higher churn.

Potential action:

Help newer members establish sustainable workout routines through introductory programs, trainer guidance, or structured workout plans.

4. Use behavioral signals and not demographics to prioritize intervention

Age, sex, and primary fitness goal showed relatively small differences in churn compared with behavioral and accessibility measures.

Potential action:

Prioritize **behavioral engagement indicators** when identifying members who may benefit from retention outreach.

Limitations

- The analysis is observational and identifies associations rather than causation.
- Churn was measured only through February, limiting long-term retention analysis.
- Linear regression was used as an exploratory approach even though churn is binary.
- Some subgroup results may be influenced by smaller group sizes.
- The dataset does not include all potential reasons for churn, such as customer satisfaction, pricing sensitivity, scheduling constraints, or individual circumstances.

Future Analysis

A future version of this project could use logistic regression or another classification model to estimate churn probability and identify the strongest predictors while controlling for multiple variables simultaneously.

Conclusion

This analysis found that gym member churn was most strongly associated with patterns of accessibility and engagement. Distance from the gym, weekly class participation, and average session length showed statistically significant relationships with observed churn, while demographic differences were comparatively modest.

The results suggest that retention strategies should focus on identifying members with low early engagement and considering accessibility barriers when designing interventions. While the analysis does not establish causation, it demonstrates how SQL, Excel, and Tableau can be used together to transform raw membership data into actionable business insights.

Tools

SQL | Excel | Tableau

Skills Demonstrated

Data Cleaning • SQL Analysis • Exploratory Data Analysis • Statistical Testing • Data Visualization • Business Intelligence • Data Storytelling